

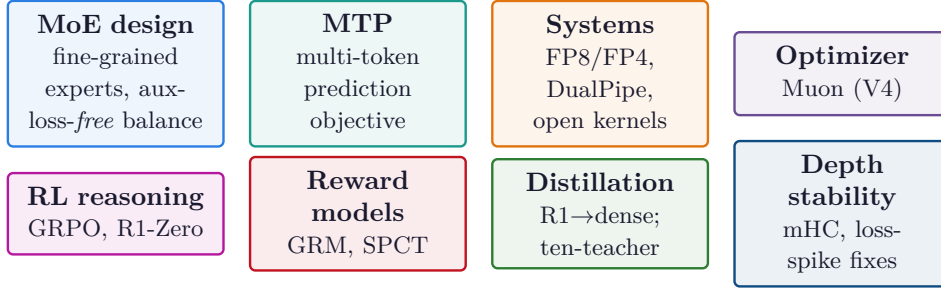
DeepSeek Beyond Attention

The other innovations — MoE, training systems, RL, reward modeling, distillation

A companion to the MLA → NSA → DSA → CSA attention report

Prepared 2026-06-14 | technical survey

The attention line (MLA→CSA) is only one of DeepSeek’s campaigns. The rest of their advantage comes from a handful of equally deliberate ideas across *six* other fronts. The same two instincts run through all of them: **relentless efficiency** (do more per FLOP, per byte, per GPU-hour) and **deletion** (the headline result is usually that some standard, complicated component was *removed*).



1 Mixture-of-Experts: specialization, then free balancing

1.1 DeepSeekMoE — fine-grained experts + a shared expert

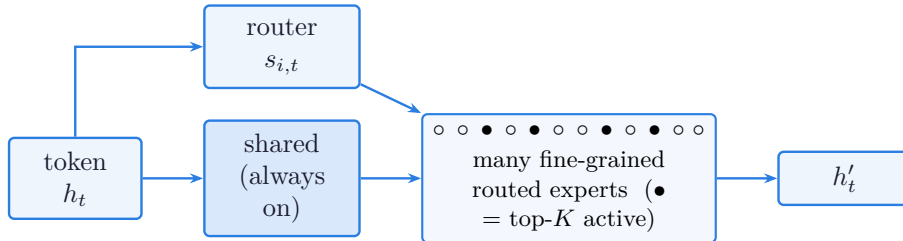
V2 / V3 / V4 backbone

Standard MoE (GShard, Switch) routes each token to one or two *large* experts. DeepSeekMoE makes two changes that raise specialization without raising cost:

- **Fine-grained segmentation:** split each expert into m smaller ones and route to mK of mN . More, smaller experts $\Rightarrow$ finer, more combinable specialization for the same active FLOPs.
- **Shared-expert isolation:** a few experts are *always on*, absorbing common/background knowledge so the routed experts need not each re-learn it.

A token’s FFN output sums the shared experts and the top- K routed experts, weighted by a gate:

$$h'_t = h_t + \sum_{i=1}^{N_s} \text{FFN}_i^{\text{shared}}(h_t) + \sum_{i=1}^{N_r} g_{i,t} \text{FFN}_i^{\text{routed}}(h_t), \quad g_{i,t} = \frac{s_{i,t} [i \in \text{Top}K]}{\sum_{j \in \text{Top}K} s_{j,t}}, \quad s_{i,t} = \sigma(h_t^\top e_i).$$



1.2 Auxiliary-loss-free load balancing — deleting the balancing loss

V3, the key MoE deletion

MoE needs experts to receive roughly equal load, or some collapse and capacity is wasted. Everyone before V3 added a *balancing loss*, which fights the language-modeling loss and slightly hurts quality. **V3 removes it.** Instead each expert i carries a scalar *bias* b_i used *only to select* the top- K (never in the gate weight), and the bias is nudged by a feedback loop:

$$\text{selection by TopK}(s_{i,t} + b_i), \quad b_i \leftarrow b_i + \gamma \text{sign}(\overline{\text{load}} - \text{load}_i).$$

Under-loaded experts get their bias raised (so they win more tokens), over-loaded ones lowered — a cheap controller, *no gradient, no quality tax*. (V4 keeps this, adding only a mild per-sequence balance term.)

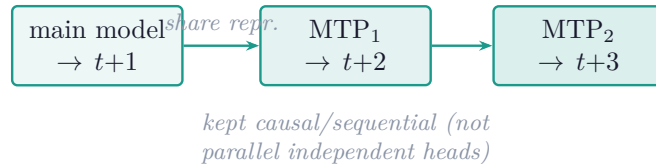
Routing constraints. To bound the all-to-all communication that MoE incurs, tokens are routed under *device-/node-limited* routing (a token’s experts span at most a few nodes), making the dispatch network-friendly. MoE sizes: V3 671B/37B-active; **V4-Pro 1.6T/49B-active**, V4-Flash 284B/13B-active.

2 Multi-Token Prediction (MTP)

V3 / V4 training objective

Next-token prediction gives one signal per position. MTP adds D lightweight *sequential* modules that each predict one token further ahead, densifying the training signal and forcing representations to plan:

$$\mathcal{L} = \mathcal{L}_{\text{next}} + \lambda \frac{1}{D} \sum_{k=1}^D \mathcal{L}_{\text{MTP}}^{(k)}, \quad \mathcal{L}_{\text{MTP}}^{(k)} = \text{CE}(\text{token}_{t+1+k} \mid \text{module}_k(h_t, \text{emb}_{t+k})).$$



The modules keep the causal chain (unlike parallel-head MTP), which trains better. **Bonus:** at inference the same modules act as a built-in draft model for *self-speculative decoding* — a training objective that doubles as a serving speedup. (Discardable if you only want the speedup gone.)

3 Training systems: low precision + overlap as co-design

V3 infra; V4 FP4

FP8 mixed-precision (V3). DeepSeek trained a frontier model with the GEMMs in **FP8** — previously considered too unstable — by quantizing *fine-grained*: per-tile (1×128) for activations, per-block (128×128) for weights, with periodic high-precision (FP32) accumulation and a few sensitive operations left in higher precision. This roughly halves compute and memory traffic vs. BF16. **V4** pushes to **FP4 quantization-aware training** on the MoE expert weights and the indexer’s QK path (FP8 elsewhere).

DualPipe (V3). A bidirectional pipeline-parallel schedule that *overlaps* forward and backward work and, crucially, hides the MoE all-to-all dispatch/combine behind computation — driving pipeline “bubbles” and communication overhead toward zero. This is what made the \$5.6M / 2,000-H800 V3 run possible.

Open infrastructure. The kernels are treated as research and open-sourced: **DeepGEMM** (FP8 GEMM), **DeepEP** (expert-parallel comms), **EPLB** (expert-parallel load balancer), **3FS** (parallel file system), **FlashMLA**, and the **Fire-Flyer AI-HPC** cluster co-design paper. The recurring thesis: *algorithm and hardware are designed together*, and the kernels are part of the contribution.

4 Optimizer: Muon (V4)

V4, replaces AdamW for matrices

V4 trains most parameters with **Muon** instead of AdamW (AdamW kept only for embeddings, the prediction head, and RMSNorm weights). Muon orthogonalizes the momentum of each 2-D weight matrix before stepping — via a few *Newton-Schulz* iterations that approximate $(MM^\top)^{-1/2}M$ — which conditions the update and gives faster, more stable convergence at scale:

$$W \leftarrow W - \eta \cdot \text{NewtonSchulz}(M), \quad \text{NewtonSchulz}(M) \approx \text{orth}(M).$$

Notably DeepSeek *drops* the QK-Clip stabilization that earlier Muon runs (Kimi K2) needed — RMSNorm on the queries and compressed-KV entries already does that job. (Another small deletion.) Trained over $\sim 32T$ (Flash) / $33T$ (Pro) tokens.

5 Reasoning from reinforcement learning

5.1 GRPO — dropping the critic

introduced in DeepSeekMath

PPO needs a value network (a second large model) to estimate advantages. **GRPO removes it.** For each prompt it samples a *group* of G answers, scores them, and uses the *group-normalized* reward as the advantage — the group average plays the role the critic used to:

$$A_i = \frac{r_i - \text{mean}(r_1, \dots, r_G)}{\text{std}(r_1, \dots, r_G)},$$

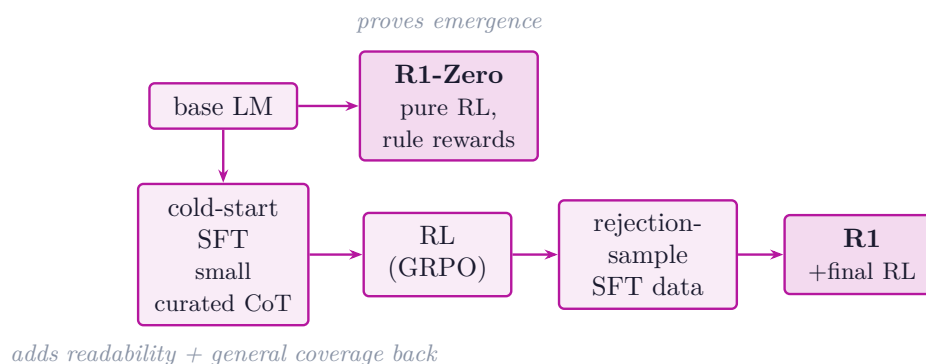
$$\mathcal{J}_{\text{GRPO}} = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_t \min(\rho_{i,t} A_i, \text{clip}(\rho_{i,t}, 1-\epsilon, 1+\epsilon) A_i) \right] - \beta \text{KL}(\pi_\theta \| \pi_{\text{ref}}), \quad \rho_{i,t} = \frac{\pi_\theta(o_{i,t})}{\pi_{\theta_{\text{old}}}(o_{i,t})}.$$

Half the RL memory disappears with the critic; the advantage is now just “better or worse than your siblings.” GRPO is now the field-default reasoning-RL algorithm.

5.2 R1-Zero / R1 — reasoning as an emergent property of RL

R1, published in Nature

R1-Zero applies GRPO to the *base* model with *no supervised fine-tuning at all*, using only **rule-based rewards**: an *accuracy* reward (is the final answer verifiably correct?) plus a *format* reward (did it put reasoning in `<think>` tags?). From this alone, long chains of thought, self-verification, and “aha” backtracking *emerge*. **R1** then adds a small amount of structure back to fix readability and broaden coverage:



The scientific claim — that reasoning need not be taught by imitation but can be *elicited* by RL — is what reset the field’s view of reasoning models.

6 Reward modeling: GRM + SPCT

April 2025

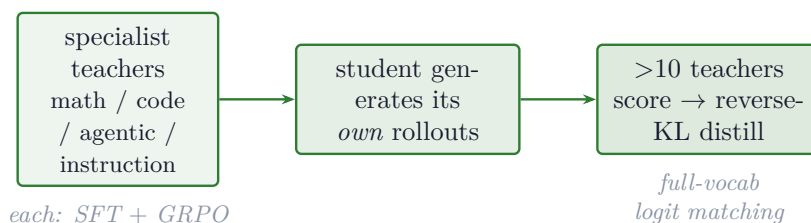
Rule-based rewards only work where answers are verifiable (math, code). For *general* domains DeepSeek-GRM trains a **generative reward model**: instead of emitting a bare scalar, the RM first *writes principles* appropriate to the query, then *critiques* the candidate against them, then scores — trained by **Self-Principled Critique Tuning (SPCT)** (rejection sampling + RL). Because the RM is generative, you can **scale it at inference**: sample many principle–critique passes and aggregate (vote) to get a better reward signal by spending more test-time compute — inference-time scaling applied to the *judge*, not just the policy.

7 Distillation

R1 era; V4 “ten teachers”

R1 → dense (2025). The reasoning of the big R1 model is distilled into small *dense* models (1.5B–70B) by plain SFT on R1’s generated reasoning traces — showing the capability transfers and that small models reason better via distillation than via their own RL.

V4 “ten teachers, one student” (2026). V4 replaces a single mixed-RL stage with multi-teacher *on-policy* distillation:



- **Teachers** are specialists, each SFT+GRPO-trained; notably the *reward model is the actor* (the specialist judges its own rubric-guided trajectories).
- **Consolidation** is on-policy (the student samples, teachers grade its samples) minimizing *reverse-KL* to the teacher mixture — and uses **full-vocabulary logit distillation** (match the entire distribution at every position, not just sampled tokens) via custom TileLang kernels that recompute logits on the fly instead of materializing them.

8 Depth & stability: mHC and loss-spike fixes (V4)

V4

To stack the heavily-compressed attention very deep, V4 replaces plain residual connections with **Manifold-Constrained Hyper-Connections (mHC)** — extending ByteDance’s Hyper-Connections by widening the residual into n parallel streams and constraining the inter-stream mixing matrix to be *doubly stochastic* (the Birkhoff polytope), projected there each step via **Sinkhorn–Knopp** (~ 20 iterations). Double stochasticity bounds the spectral norm, so signal is mixed but never amplified — keeping very deep stacks stable:

$$H_{\ell+1} = A_{\ell}H_{\ell} + (\text{layer output}), \quad A_{\ell}\mathbf{1} = \mathbf{1}, \quad A_{\ell}^{\top}\mathbf{1} = \mathbf{1}, \quad A_{\ell} \geq 0.$$

Reported alongside are training-time *loss-spike* fixes (*Anticipatory Routing*, *SwiGLU Clamping*). (*V4 internals here are from third-party summaries of the tech report, not fully official.*)

9 The playbook, in one place

Technique	First in	The move (and what it deletes)
DeepSeekMoE	V2	fine-grained + shared experts — better specialization, same FLOPs
Aux-loss-free balance	V3	bias-feedback routing — <i>deletes the balancing loss</i>
Multi-Token Prediction	V3	denser signal; <i>free</i> self-speculative decoding
FP8 / FP4 training	V3 / V4	fine-grained low-precision GEMMs — halve (then quarter) compute
DualPipe + open kernels	V3	overlap comms behind compute — <i>deletes pipeline bubbles</i>
Muon	V4	orthogonalized updates; <i>deletes QK-Clip</i>
GRPO	DeepSeekMath	group-relative advantage — <i>deletes the critic</i>
R1-Zero / R1	2025	reasoning emerges from pure RL — <i>deletes the SFT requirement</i>
GRM / SPCT	2025	generative, inference-scalable reward model
R1→dense / ten-teacher	2025 / V4	distill big reasoning into small/uni models
mHC	V4	doubly-stochastic residual mixing for deep stability

Read top to bottom and the signature is unmistakable: nearly every row is a **deletion** — of a loss, a network, a stage, a stabilizer, a precision tier, a communication stall. DeepSeek’s edge is less a pile of new mechanisms than a discipline of finding the complicated thing everyone else kept, showing it was unnecessary, and *co-designing* the cheap replacement down to the GPU kernel.

Sources: DeepSeekMoE, DeepSeek-V2/V3 reports, DeepSeekMath (GRPO), DeepSeek-R1 (Nature 2025), DeepSeek-GRM (arXiv:2504.02495), and the V3 systems/Fire-Flyer papers are primary. V4-era items (Muon configuration, mHC / Sinkhorn–Knopp, “ten teachers” distillation, FP4 QAT, loss-spike fixes, MoE sizes) are drawn from third-party summaries of the V4 technical report (Apr–May 2026) and are reported as published, not re-derived.